

Q1. ^{Given} $2x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (x^2 - 3)y = 0$

Soln) here $x=0$ is a regular singular point.

~~now~~ for finding indicial eqⁿ
compare the eqⁿ with

$$x^2 y'' + x p(x) y' + q(x) y = 0$$

then

$$p(x) = \frac{1}{2}$$

$$q(x) = \frac{x^2 - 3}{2}$$

$$q_0 = q(0) = -\frac{3}{2}$$

∴ indicial eqⁿ is given by

$$r(r-1) + p_0 r + q_0 = 0$$

$$\Rightarrow r(r-1) + \frac{1}{2}r + \left(-\frac{3}{2}\right) = 0$$

$$\Rightarrow r^2 - r + \frac{r}{2} - \frac{3}{2} = 0$$

$$\Rightarrow r^2 - \frac{r}{2} - \frac{3}{2} = 0$$

$$\Rightarrow 2r^2 - r - 3 = 0$$

$\therefore$ roots are $\mu_1 = \frac{3}{2}$ $\mu_2 = -1$

or

assume $y = x^\mu \sum_{n=0}^{\infty} C_n x^n$

where $C_0 \neq 0$

$$y' = \sum_{n=0}^{\infty} (n+\mu) C_n x^{n+\mu-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+\mu)(n+\mu-1) C_n x^{n+\mu-2}$$

putting in the diffⁿ eqⁿ

$$2 \sum_{n=0}^{\infty} (n+\mu)(n+\mu-1) C_n x^{n+\mu}$$

$$+ \sum_{n=0}^{\infty} (n+\mu) C_n x^{n+\mu} + \sum_{n=0}^{\infty} C_n x^{n+\mu+2}$$

$$- 3 \sum_{n=0}^{\infty} C_n x^{n+\mu} = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} [2(n+\mu)(n+\mu-1) + (n+\mu) - 3] C_n x^{n+\mu}$$

$$+ \sum_{n=2}^{\infty} C_{n-2} x^{n+\mu} = 0$$

$$\Rightarrow [2\mu(\mu-1) + \mu - 3] C_0 x^\mu$$

$$+ [2(\mu+1)\mu + (\mu+1) - 3] C_1 x^{\mu+1}$$

$$+ \sum_{n=2}^{\infty} \{ [2(n+\mu)(n+\mu-1) + (n+\mu) - 3] C_n + C_{n-2} \} x^{n+\mu} = 0$$

Equating to zero the coeff of lowest power of x , we obtain indicial eqⁿ.

$$2x(x-1) + x - 3 = 0$$

$$2x^2 - x - 3 = 0$$

$$x_1 = \frac{3}{2}, \quad x_2 = -1$$

Equating to zero the coeff of higher power of x

$$[2(x+1)x + (x+1) - 3]c_1 = 0 \quad \text{--- (1)}$$

and the recurrence formula =

$$[2(n+x)(n+x-1) + (n+x) - 3]c_n + c_{n-2} = 0$$

; $n \geq 2$ --- (2)

assume $x = x_1 = \frac{3}{2}$

from (1)

$$\left[2 \left(\frac{5}{2} \right) \cdot \frac{3}{2} + \frac{5}{2} - 3 \right] c_1 = 0$$

$$7c_1 = 0$$

$$\Rightarrow c_1 = 0$$

from (2)

$$\left[2 \binom{n+3}{2} \binom{n+1}{2} + \binom{n+3}{2} - 3 \right] c_n + c_{n-2} = 0$$

$$\left[\frac{2(2n+3)(2n+1)}{2} + \frac{(2n+3)}{2} - 3 \right] c_n + c_{n-2} = 0$$

$$\left[(2n+3)(2n+1) + (2n+3) - 6 \right] c_n + 2c_{n-2} = 0$$

$$\left[4n^2 + 10n + 3 + 2n + 3 - 6 \right] c_n + 2c_{n-2} = 0$$

$$\left[4n^2 + 10n \right] c_n + 2c_{n-2} = 0$$

$$\left[2n^2 + 5n \right] c_n + c_{n-2} = 0$$

$$\left[n(2n+5) \right] c_n + c_{n-2} = 0$$

$$c_n = - \frac{c_{n-2}}{n(2n+5)}, n \geq 2$$

$$c_2 = - \frac{c_0}{18} \quad c_3 = - \frac{c_1}{33} = 0 \quad \therefore c_1 = 0$$

$$c_4 = - \frac{c_2}{52} = \frac{c_0}{936}$$

all the odd coeff are zero

$$\therefore y_1 = x^{\frac{3}{2}} \sum_{n=0}^{\infty} a_n x^n$$

$$\text{for } x = \frac{3}{2}$$

$$y_1 = x^{3/2} [c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots]$$

$$= x^{3/2} \left[c_0 - \frac{c_0}{18} x^2 + \frac{c_0}{936} x^4 + \dots \right]$$

$$y_1 = c_0 x^{3/2} \left[1 - \frac{1}{18} x^2 + \frac{x^4}{936} + \dots \right]$$

assume $x_1 = x_2 = -1$

from (1)

$$[2(-1+1)(-1) + (-1+1) - 3] c_1 = 0$$
$$\Rightarrow c_1 = 0$$

from (2)

$$[2(n-1)(n-1-1) + (n-1) - 3] c_n + c_{n-2} = 0$$

$$[2(n-1)(n-2) + (n-1) - 3] c_n + c_{n-2} = 0$$

$$[2n^2 - 6n + 4 + n - 1 - 3] c_n + c_{n-2} = 0$$

$$[2n^2 - 5n] c_n + c_{n-2} = 0$$

$$n(2n-5)c_n + c_{n-2} = 0$$

$$c_n = -\frac{c_{n-2}}{n(2n-5)} \quad n \geq 2$$

$$c_2 = \frac{c_0}{2}, \quad c_3 = -\frac{c_1}{3} = 0 \quad \because c_1 = 0$$

$$c_4 = -\frac{c_2}{12} = -\frac{1}{12} \left(\frac{c_0}{2} \right) = -\frac{c_0}{24}$$

all the odd coeff are zero.

$$y_2 = x^{-1} [c_0 x + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots]$$

$$y_2 = x^{-1} \left[c_0 + \frac{c_0 x^2}{2} - \frac{c_0 x^4}{24} + \dots \right]$$

$$= c_0 x^{-1} \left[1 + \frac{1}{2} x^2 - \frac{1}{24} x^4 + \dots \right]$$

$\therefore y_1$ and y_2 correspond to roots $x_1 = \frac{3}{2}$

and $x_2 = -1$

y_1 and y_2 are I. I
General solution is given by

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

where c_1 and c_2 are arb. constants
and y_1 and y_2 are series solution given above.

Q (2) Solve the I.V.P using Laplace Transform

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 5y = h(t)$$

$$\text{where } h(t) = \begin{cases} 1, & 0 \leq t < \pi \\ 0, & t \geq \pi \end{cases}$$

$$y(0) = 0$$

$$y'(0) = 0$$

Soln

Taking Laplace Transform on both sides of the D.E

$$\mathcal{L}\{y''\} = s^2 Y(s) - sy(0) - y'(0) = s^2 Y(s)$$

$$\mathcal{L}\{y'\} = sY(s) - y(0) = sY(s)$$

$$\mathcal{L}\{h(t)\} = \int_0^{\infty} e^{-st} h(t) dt$$

$$= \int_0^{\pi} e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_0^{\pi} = \frac{1 - e^{-\pi s}}{s}$$

thus from D.E

$$y'' + 2y' + 5y = h(t) \quad \text{we get}$$

$$\Rightarrow [s^2 + 2s + 5] Y(s) = \frac{1 - e^{-\pi s}}{s}$$

$$Y(s) = \frac{1 - e^{-\pi s}}{s(s^2 + 2s + 5)}$$

taking inverse Laplace transform

$$\mathcal{L}^{-1} \left\{ \frac{1 - e^{-\pi s}}{s(s^2 + 2s + 5)} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{s(s^2 + 2s + 5)} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-\pi s}}{s(s^2 + 2s + 5)} \right\}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s(s^2 + 2s + 5)} \right\} = \frac{1}{5} \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\}$$

$$- \frac{1}{5} \mathcal{L}^{-1} \left\{ \frac{s+2}{(s+1)^2 + 4} \right\}$$

$$= \frac{1}{5} \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - \frac{1}{5} \mathcal{L}^{-1} \left\{ \frac{s+1}{(s+1)^2 + 4} \right\}$$

$$- \frac{1}{10} \mathcal{L}^{-1} \left\{ \frac{2}{(s+1)^2 + 4} \right\}$$

$$= \frac{1}{5} - \frac{1}{5} e^{-t} \cos 2t - \frac{1}{10} e^{-t} \sin 2t$$

$$\text{let } F(s) = \frac{1}{s(s^2 + 2s + 5)}$$

$$\text{and } f(t) = \frac{1}{5} - \frac{1}{5} e^{-t} \cos 2t - \frac{1}{10} e^{-t} \sin 2t$$

$$\text{thus } \mathcal{L}^{-1}\{F(s)\} = f(t)$$

$$\mathcal{L}^{-1} \left\{ \frac{e^{-\pi s}}{s(s^2 + 2s + 5)} \right\} = \mathcal{L}^{-1}\{e^{-\pi s} F(s)\}$$

(Using Second Shifting Theorem

$$\mathcal{L}^{-1}\{e^{-\pi s} f(s)\} = U(t-\pi) f(t-\pi)$$

$$U(t-\pi) f(t-\pi) = \begin{cases} 0 & , 0 < t < \pi \\ f(t-\pi), & t > \pi \end{cases}$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s(s^2+2s+5)}\right\}$$

$$= \begin{cases} 0 & , 0 < t < \pi \\ \frac{1}{5} - \frac{1}{5} e^{-(t-\pi)} \cos 2(t-\pi) \\ - \frac{1}{10} e^{-(t-\pi)} \sin 2(t-\pi), & t > \pi \end{cases}$$

$$= \begin{cases} 0 & , 0 < t < \pi \\ \frac{1}{5} - \frac{1}{5} e^{-(t-\pi)} \cos 2t - \frac{1}{10} e^{-(t-\pi)} \sin 2t, & t > \pi \end{cases}$$

Final solution

$$y = \mathcal{L}^{-1}\left\{\frac{1 - e^{-\pi s}}{s(s^2+2s+5)}\right\}$$

$$\begin{cases} \frac{1}{5} - \frac{1}{5} e^{-t} \cos 2t - \frac{1}{10} e^{-t} \sin 2t - 0, & 0 < t < \pi \\ \frac{1}{5} - \frac{1}{5} e^{-t} \cos 2t - \frac{1}{10} e^{-t} \sin 2t - \frac{1}{5} + \frac{1}{5} e^{-(t-\pi)} \\ \cos 2t + \frac{1}{10} e^{-(t-\pi)} \sin 2t, & t > \pi. \end{cases}$$

①

$$y(t) = \begin{cases} \frac{1}{5} [1 - e^{-t} (\cos 2t + \sin 2t)]; & 0 < t < \pi \\ \frac{e^{-t}}{5} [(e^{\pi} - 1) \cos 2t + (e^{\pi} - 1) \sin 2t]; & t > \pi \end{cases}$$

(Ans)

Q-3 Find the inverse Laplace transform

1) $\mathcal{L}^{-1} \left\{ \tan^{-1} \left(\frac{1}{s} \right) \right\}$

2) $\mathcal{L}^{-1} \left\{ \frac{1}{s} \ln \left(\frac{s+a}{s+b} \right) \right\}$

Soln 1) Let $F(s) = \tan^{-1} \left(\frac{1}{s} \right)$

Diff w.r.t. 's'

①
$$F'(s) = \frac{1}{1 + \left(\frac{1}{s} \right)^2} \left(-\frac{1}{s^2} \right)$$
$$= -\frac{1}{s^2 + 1}$$

① Use $\mathcal{L}^{-1} \{ F'(s) \} = -t f(t)$
So,

$$-t f(t) = \mathcal{L}^{-1} \left\{ -\frac{1}{s^2 + 1} \right\}$$

$$\Rightarrow -t f(t) = -\sin t$$
$$f(t) = \frac{\sin t}{t}$$

①
$$\mathcal{L}^{-1} \left\{ \tan^{-1} \left(\frac{1}{s} \right) \right\} = \frac{\sin t}{t} \quad (\text{Ans})$$

$$2) \mathcal{L}^{-1} \left\{ \frac{1}{s} \ln \left(\frac{s+a}{s+b} \right) \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{\ln \left(\frac{s+a}{s+b} \right)}{s} \right\}$$

① Use $\mathcal{L} \left\{ \int_0^t f(u) du \right\} = \frac{F(s)}{s}$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{F(s)}{s} \right\} = \int_0^t f(u) du$$

$$F(s) = \ln \left(\frac{s+a}{s+b} \right) = \ln(s+a) - \ln(s+b)$$

$$f(u) = \mathcal{L}^{-1} \{ F(s) \}$$

$$= \mathcal{L}^{-1} \{ \ln(s+a) - \ln(s+b) \}$$

$$= \mathcal{L}^{-1} \{ \ln(s+a) \} - \mathcal{L}^{-1} \{ \ln(s+b) \}$$

$$\mathcal{L}^{-1} \{ \ln(s+a) \}$$

$$\because \mathcal{L}^{-1} \left\{ \frac{1}{s+a} \right\} = e^{-at}$$

$$\Rightarrow \mathcal{L}^{-1} \{ \ln(s+a) \} = \mathcal{L}^{-1} \{ F'(s) \} = -t f(t)$$

$$-t f(t) = e^{-at}$$

$$f(t) = -\frac{1}{t} e^{-at}$$

$$f(u) = \mathcal{L}^{-1} \{ \ln(s+a) \} - \mathcal{L}^{-1} \{ \ln(s+b) \}$$

$$= -\frac{1}{u} [e^{-au} - e^{-bu}]$$

① $\left\{ \mathcal{L}^{-1} \left\{ \frac{1}{s} \ln \left(\frac{s+a}{s+b} \right) \right\} = \int_0^t \frac{e^{-bu} - e^{-au}}{u} du \right.$

(Ans)

Q-4. Show that $\int x^2 J_0(x) dx = x^2 J_1(x) + x J_0(x) - \int J_0(x) dx$

Proof -

$$\because [x^0 J_0(x)]' = x^0 J_{0-1}(x)$$

$$[x J_1(x)]' = x J_0(x)$$

multiply by x both sides

$$x^2 J_0(x) = x \frac{d}{dx} (x J_1(x))$$

integrating both sides

$$\int x^2 J_0(x) dx = \int x \frac{d}{dx} (x J_1(x)) dx$$

suppose $v = x J_1(x)$

$$dv = \frac{d}{dx} (x J_1(x)) dx$$

$$\int x \frac{d}{dx} (x J_1(x)) dx$$

$$= x(x J_1(x)) - \int x J_1(x) dx$$

$$\int x^2 J_0(x) dx = x^2 J_1(x) - \int x J_1(x) dx$$

now

$$\frac{d}{dx} (x J_0(x)) = J_0(x) + x(J_0(x))'$$

using $[x^{-0} J_0(x)]' = -x^{-0} J_{0+1}(x)$

$$[J_0(x)]' = -J_1(x)$$

$$\frac{d}{dx} (x J_0(x)) = J_0(x) - x J_1(x)$$

$$x J_1(x) = J_0(x) - \frac{d}{dx} (x J_0(x))$$

integrating both sides

$$\int x J_1(x) dx = \int J_0(x) dx - x J_0(x)$$

②

$$\therefore \int x^2 J_0(x) dx = x^2 J_1(x) + x J_0(x) - \int J_0(x) dx$$

$$\boxed{\int x^2 J_0(x) dx = x^2 J_1(x) + x J_0(x) - \int J_0(x) dx}$$

another way

②

using the identity

$$J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_{\nu}(x)$$

put $\nu = 0$

$$J_{-1}(x) + J_1(x) = \frac{2 \cdot 0}{x} J_0(x)$$

$$\Rightarrow J_{-1}(x) = -J_1(x)$$

$$\frac{d}{dx} (x^{\nu} J_{\nu}(x)) = x^{\nu} J_{\nu-1}(x)$$

put $\nu = 2$

$$\frac{d}{dx} (x^2 J_2(x)) = x^2 J_1(x)$$

Start from

$$x^2 J_0(x) = x (x J_0(x))$$

$$\frac{d}{dx} (x J_1(x)) = x J_0(x)$$

$$x^2 J_0(x) = x \left(\frac{d}{dx} (x J_1(x)) \right)$$

integrating

$$\int x^2 J_0 dx = x^2 J_1(x) - \int x J_1(x) dx \quad (1)$$

$$\frac{d}{dx} (x J_0(x)) = J_0(x) + x \frac{d}{dx} (J_0(x))$$

$$\text{using } \frac{d}{dx} (x^{-\nu} J_\nu(x)) = -x^{-\nu} J_{\nu+1}(x)$$

$$\Rightarrow \frac{d}{dx} (J_0(x)) = -J_1(x)$$

$$\frac{d}{dx} (x J_0(x)) = J_0(x) - x J_1(x)$$

$$x J_1(x) = J_0(x) - \frac{d}{dx} (x J_0(x))$$

putting in (1)

$$\int x^2 J_0 dx = x^2 J_1(x) - \int J_0(x) dx$$

$$\int x^2 J_0 dx = x^2 J_1(x) + x J_0(x) - \int J_0(x) dx$$